

Concurrent Guillain-Barré Syndrome and Acute Disseminated Encephalomyelitis in a Child: A Case Report from Bangladesh

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ABSTRACT

Guillain-Barré syndrome (GBS) and acute disseminated encephalomyelitis (ADEM) are postinfectious, inflammatory, and neurological disorders involving the peripheral and central nervous system, respectively. Although the clinical features are distinct, they share a common pathogenesis. Both the disorders are not uncommon in children. Concurrent GBS and ADEM occurring simultaneously is a rare entity. We are reporting a case of a 5-year-old boy presenting with acute-onset ascending paralysis with irritability, hyperreflexia, and positive Babinski sign. This patient was diagnosed as having both GBS and ADEM. The CSF albumin-cytological disassociation, electrophysiological diagnosis, and MRI of brain supported the concurrent present of the two disorders.

KEYWORDS: *Acute disseminated encephalomyelitis, concurrent, Guillain-Barré syndrome*

INTRODUCTION

Guillain-Barré syndrome (GBS) is the most common cause of acute flaccid paralysis in healthy infants and children. It is an acute postinfectious inflammatory disorder of the peripheral nervous system (PNS).^[1-3] Acute disseminated encephalomyelitis (ADEM) is an immunologically mediated inflammatory disease of the central nervous system (CNS), resulting in multifocal demyelinating lesions affecting the gray and white matter of the brain and spinal cord. Most of the patients experience a combination of neurological features depending on the site of the lesion. The key clinical feature is encephalopathy, varying from lethargy to coma along with upper motor neuron lesion features, ataxia, hemiparesis, seizure, vision loss, speech impairment, and spinal cord involvement.^[4-6]

GBS and ADEM are recognized as separate entities, yet they share some common features such as autoimmune pathogenesis, injury of the myelin tissue, history of infection or vaccination, etc. Treatment modalities are

different, so the distinction of these two disorders and the selection of appropriate treatment is vital.^[7] Here, we report a case of a 5-year-old boy having involvement of both CNS and PNS and who was diagnosed as having concurrent GBS and ADEM.

CASE REPORT

A 5-year-old boy presented with weakness of both limbs for 2 days. He also complained of severe pain in both lower limbs along with irritability. Gradually, weakness was progressive; he developed difficulty in walking and was unable to walk for the next 2 days. Meanwhile, he also developed weakness of both upper limbs. His bowel and bladder habits were normal. He had no history of seizure, loss of consciousness, altered pattern of behavior except restlessness, vomiting, neck pain, visual or hearing impairment, any history of taking drugs, or a history of recent vaccination. On

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query, he had a history of varicella infection three weeks ago. He was born by lower uterine caesarian section, there was no perinatal adverse event, and he was born from nonconsanguineous parents. None of the family members had a history of immune disorders or a similar type of illness. He had a normal development before this illness.

On examination, the child was irritable and conscious; vitals were within normal limit, but there was a cold periphery. He was anicteric and not pale, and signs of meningeal irritation were absent. Anthropometry was within normal range. On nervous system examination, higher psychic function was normal; the patient was conscious, cooperative but restless. The bulk of both upper and lower limbs were normal, tone reduced in both upper and lower limbs, Power was 1/5 in all four limbs, knee and ankle jerks were exaggerated, planter was extensor. In the upper limb, all the jerks were normal. There was no involuntary movement, and sensory nerve examination revealed that there was hyperesthesia. Gait could not be evaluated, and cerebellar function was intact as far as examined. Fundoscopic examination revealed no abnormality.

Investigations showed: Complete blood count: HB% 13%, WBC count: 11,000, neutrophil: 56%, lymphocyte:

39%, ESR: normal, and platelet count: 2,00,000. CRP, electrolytes, and serum calcium were normal. CSF study revealed cell count 2, all lymphocytes, glucose was 40 mg/dL, and protein was 90 mg/dL. The CSF study was done during the second week of the disease, which showed an albuminocytological dissociation.

Nerve conduction velocity (NCV) and electromyography (EMG): NCS was done during the third week of the disease. Sensory nerve action potential and conduction velocity were normal. Compound muscle action potential (CMAP) was absent in median, ulnar, and peroneal nerves and posterior tibial nerves; motor conduction velocity and distal motor latency were normal. F-waves were absent in median and ulnar nerves. An EMG showed positive sharp waves and fibrillations, which were suggestive of acute motor axonal polyneuropathy (AMAN). An MRI of the brain showed: hyperintensity in multiple areas of the brain, namely basal ganglia; hyperintensity of the gray-white interface in bilateral parietal and temporal regions, suggestive of ADEM [Figure 1]. An MRI of the spine was normal.

We started treatment with intravenous immunoglobulin 400 mg/kg/day for 5 days; subsequently, intravenous methyl prednisolone 30 mg/kg/day was given for

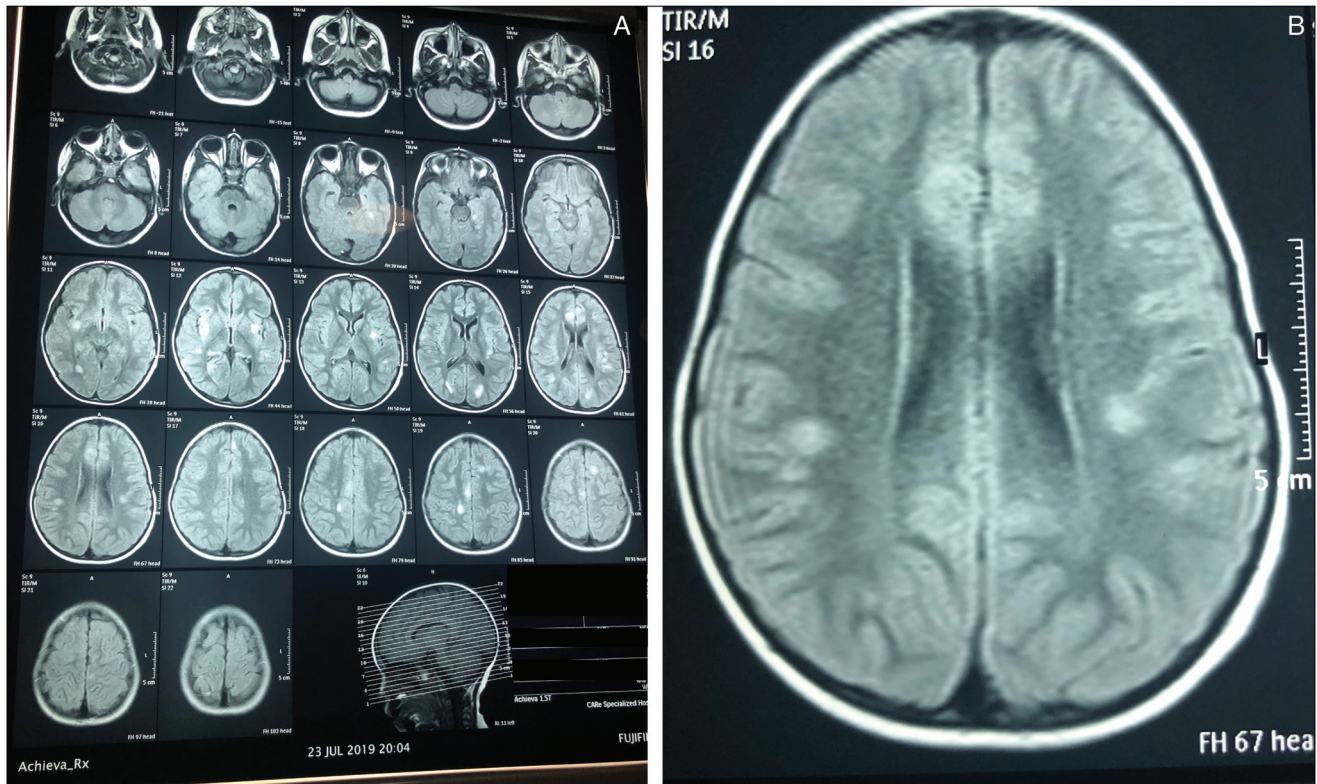


Figure 1: (A) MRI of the brain (FLAIR image) showing multiple hyperintense lesions in different parts of the brain. (B) MRI of the brain showing hyperintense lesions in the subcortical regions

5 days. There was stabilization of the clinical course of the child after the treatment and gradually there was improvement. At the 1-year follow-up, the child was completely normal with no residual deficit.

Based on clinical features, CSF study, and neurophysiological study, a diagnosis of GBS was made; based on MRI brain findings along with clinical features of upper motor neuron lesion, a diagnosis of ADEM was made.

DISCUSSION

Both GBS and ADEM are postinfectious, inflammatory, and neurological disorders, although the site of involvement is different. In the first one, there is usually involvement of the peripheral nerves; however, in the second one, there is involvement of the brain and spinal cord.^[2,8] The basic mechanism is similar in both the disorders, which is molecular mimicry and cross-reactive immune response. Eventually, there is destruction of myelin or the nerve tissue.^[7,9] However, the clinical features are distinct as well as the management is different. Thus, the diagnosis and distinction of these two entities is crucial for proper management. Very few cases have been reported in the pediatric population with both the entities.

The presenting features of our case are classical of those of GBS. The boy had a history of varicella infection followed by a progressive ascending pattern of weakness along with limb pain. He had no CNS manifestations except irritability, which could be explained by the pain. However, on examination, there was hypotonia with hyperreflexia and with positive Babinski sign. Thus, the possibility of ADEM emerged.

GBS is a postinfectious inflammatory disease of the PNS causing acute flaccid paralysis. Patients usually present with weakness starting from the lower limbs, ascending in a pattern with or without sensory features. The clinical presentation of GBS is heterogeneous, and several distinct clinical variants exist. The diagnosis is based on history, neuro-electrophysiology, MRI of the spine, and CSF albuminocytological dissociation.^[10-12] GBS has been reclassified according to clinical and electrophysiological properties. According to that, the large spectrum varies from classic acute inflammatory demyelinating polyneuropathy (AIDP) and the axonal forms, which include acute motor and sensory axonal neuropathy (AMSAN) and AMAN, to several atypical clinical types (i.e., Miller Fisher syndrome, Bickerstaff brainstem encephalitis, pharyngo-cervical-brachial variant, polyneuritis cranialis, acute sensory neuropathy, and pandysautonomia).^[13-15] Our patient

was diagnosed as an AMAN variety according to the electrophysiology. The albuminocytological dissociation was also in favor of diagnosis.

On the other hand, ADEM is an immune-mediated demyelinating CNS disorder. Its classic features are new-onset polyfocal neurologic symptoms, including encephalopathy. Neuroimaging is very characteristic, and it shows multifocal demyelination. There is a temporal relationship of ADEM with infection and vaccines. Most of the cases are monophasic.^[16-18] Our case did not have the typical features of ADEM-like seizure, loss of consciousness, or visual impairment. The features that were consistent with ADEM were irritability, hyperreflexia, and monophasic illness. However, the most important part of the diagnosis was the typical MRI finding showing multifocal subcortical lesions in the brain.

Simultaneous GBS and ADEM in children is a rare entity. Patients may present with typical features of GBS; then, they may develop ADEM and vice versa. Patients may also present with transverse myelitis along with GBS.^[19] Although autoimmunity to myelin protein antigen and nerve tissue plays a key role here, some authors suggest that there may be a continuous clinical spectrum due to antibody-mediated postinfectious syndrome, where there is the involvement of both CNS and PNS.^[20,21] Although some authors suggest a different opinion, according to them immune response against a component of myelin of the CNS may carry cross-antigenicity with the peripheral nerve. The high number of helper inducer T cells suggests a common pathogenic mechanism of GBS and ADEM.^[3]

The main challenge of this comorbid condition is the different treatment regime of the two disorders. In GBS, the first line of treatment is intravenous immunoglobulin; whereas in the case of ADEM, bolus methylprednisolone is the mainstay of treatment. However, the use of corticosteroids in GBS is disappointing. The exact mechanism is not known; the proposed mechanism is that steroids reduce macrophage infiltration and, thus, delay the regeneration of injured nerves.^[22] We started the treatment of the patient with intravenous immunoglobulin for five days; then we started intravenous methylprednisolone, which caused dramatic improvement in the child. During the one-year follow-up, there was no residual deficit in the child.

CONCLUSION

The coexistence of GBS and ADEM is a rare entity, with the involvement of the CNS and PNS. Early diagnosis and treatment is very important, as it hastens

the recovery. We report the case of a child with both GBS and ADEM, which may be the first case reported in Bangladesh.

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Conflicts of interest

There are no conflicts of interest.

Author contribution

Kanij Fatema: Preparation of article, design and final revision, and correspondence. Md Mizanur Rahman: Conceptualization, overall supervision. Shaheen Akhter: Article writing.

Ethical approval/patient consent

Ethical approval has been obtained from the institutional review board, and consent of the guardian of the patient has been obtained.

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